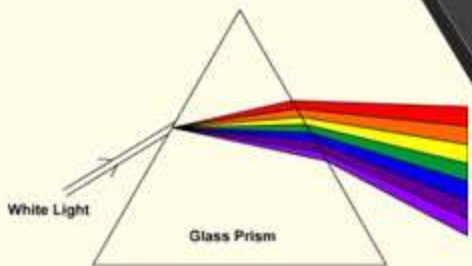


PHYSICS

GRAPHICAL REPRESENTATION OF MOTION

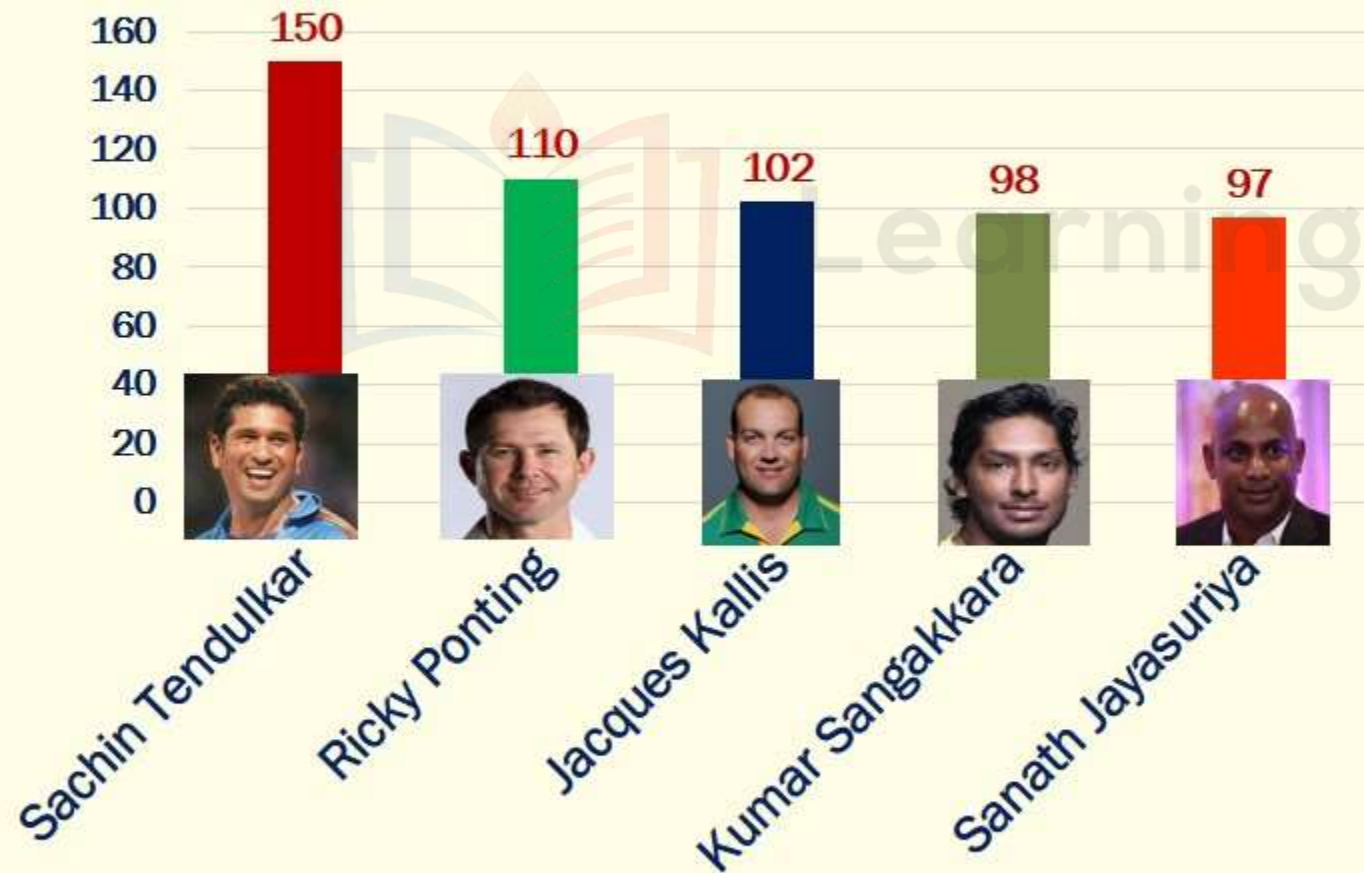
MODULE - 4



LOOK AT THE GRAPH ...WHAT DOES IT CONVEY?

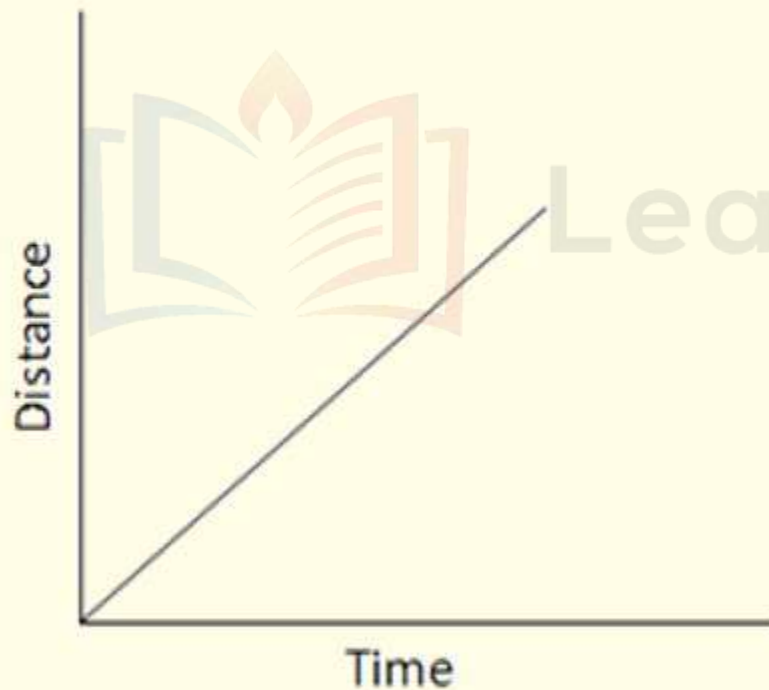


WHY WE NEED GRAPHICAL REPRESENTATION ANYWAY ?



- In any ODI or Test Match we see cricket scores being represented as graphs.
- Graphs helps us to understand a variety of information in a single paper.
- Thus, to describe the motion of an object also we can use graphs.

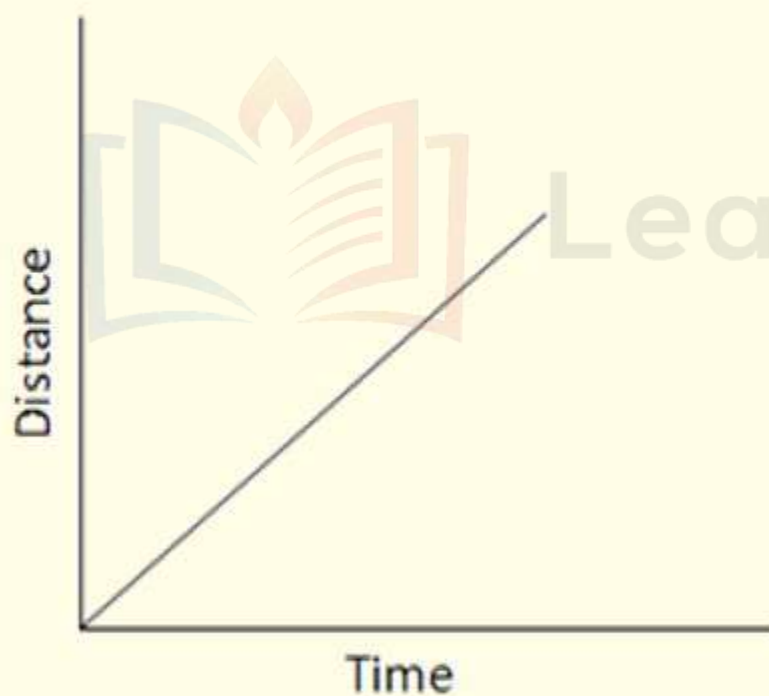
REPRESENTING MOTION BY GRAPH



- A graph is a pictorial representation of the relation between two sets of data of which **one set is of dependent variables and the other set is of independent variables.**
- To describe the motion of an object, we can use line graphs. In this case, line graphs show dependence of one physical quantity, such as distance or velocity, on another quantity, such as time.

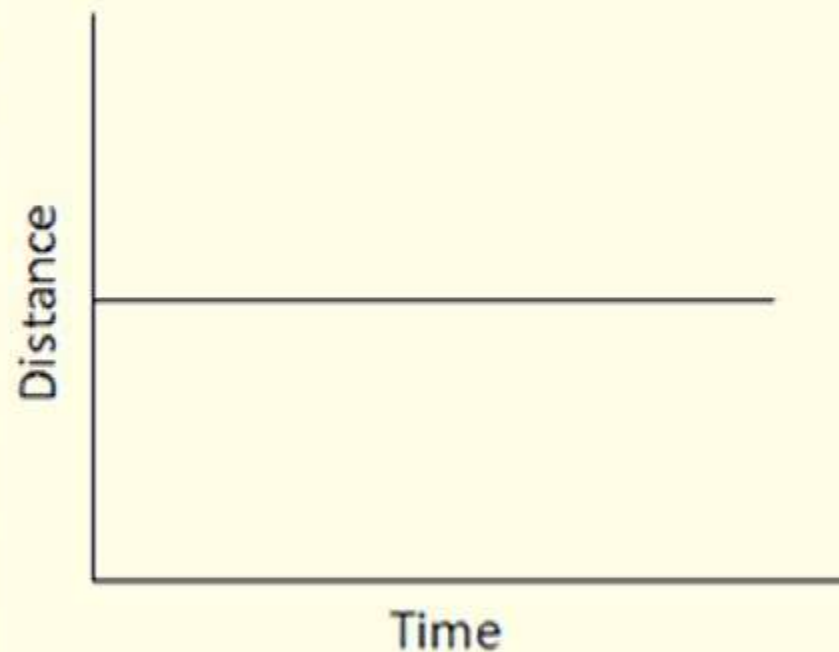


DISTANCE – TIME GRAPH



- The change in the position of an object with time can be represented on the distance-time graph.
- In this graph, time is taken along the x-axis and distance is taken along the y-axis.
- Distance time graphs of a moving body can be used to calculate the speed of the body as they specifically represent velocity.

WHAT WILL BE THE GRAPH OF A PARKED CAR?

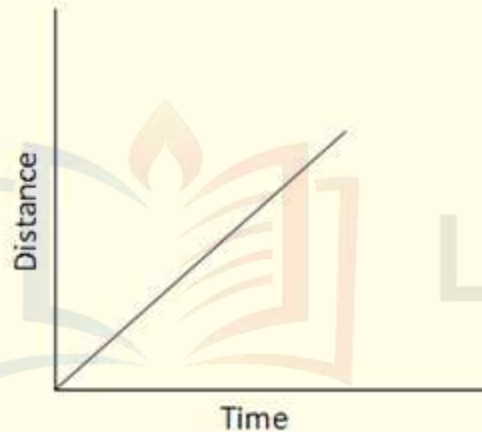


Any object at rest will have no distance travelled with change of time
thus a straight line is obtained parallel to x axis.

UNIFORM MOTION



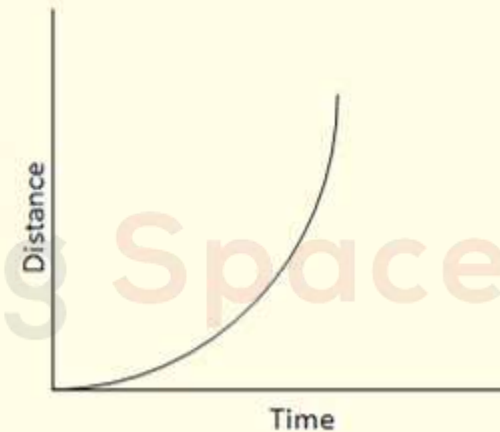
CAR ON A HIGHWAY



NON-UNIFORM MOTION

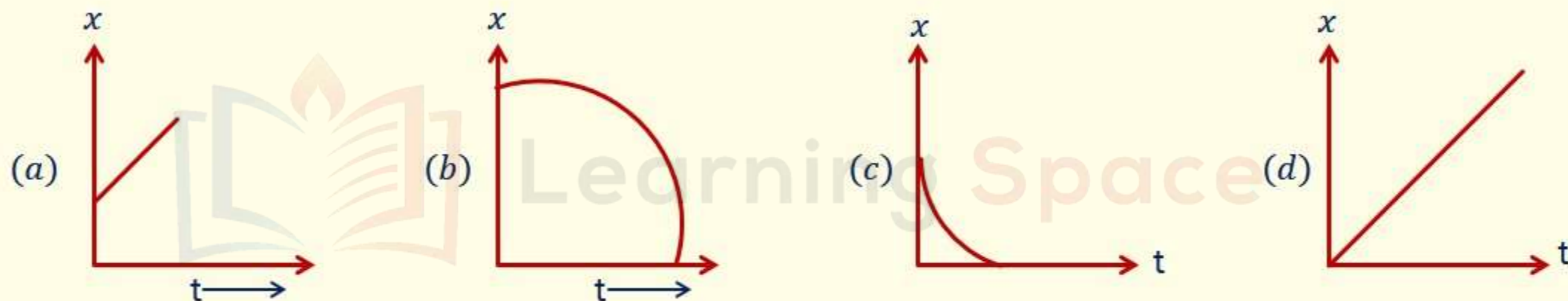


CARS ON A BUSY STREET



QUESTION 1

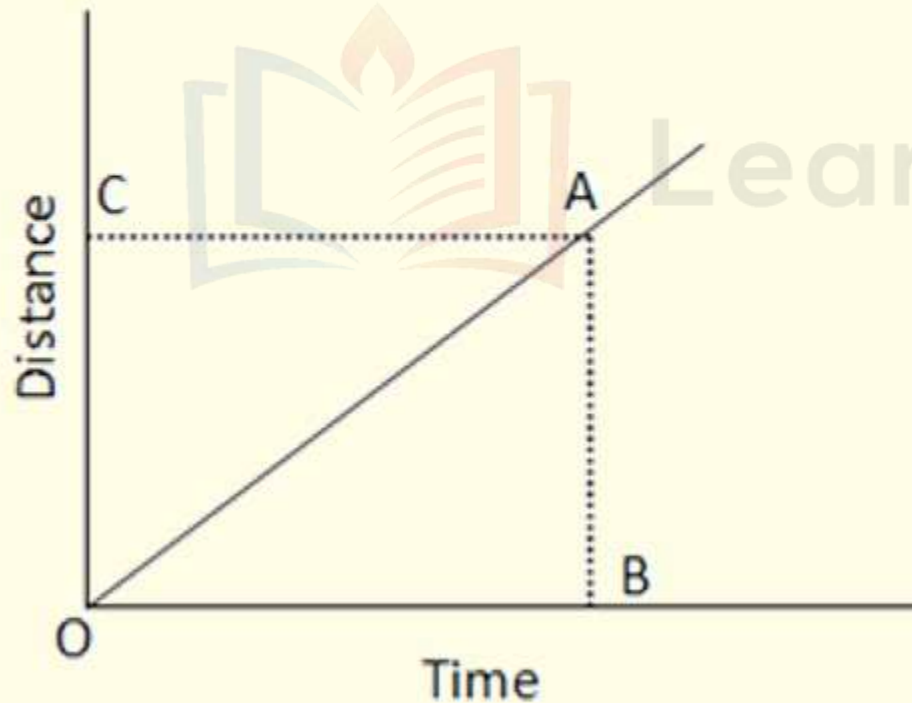
Which of the following distance time graph ($x-t$) represents one dimensional uniform motion ?





CALCULATING SPEED FROM DISTANCE - TIME GRAPH

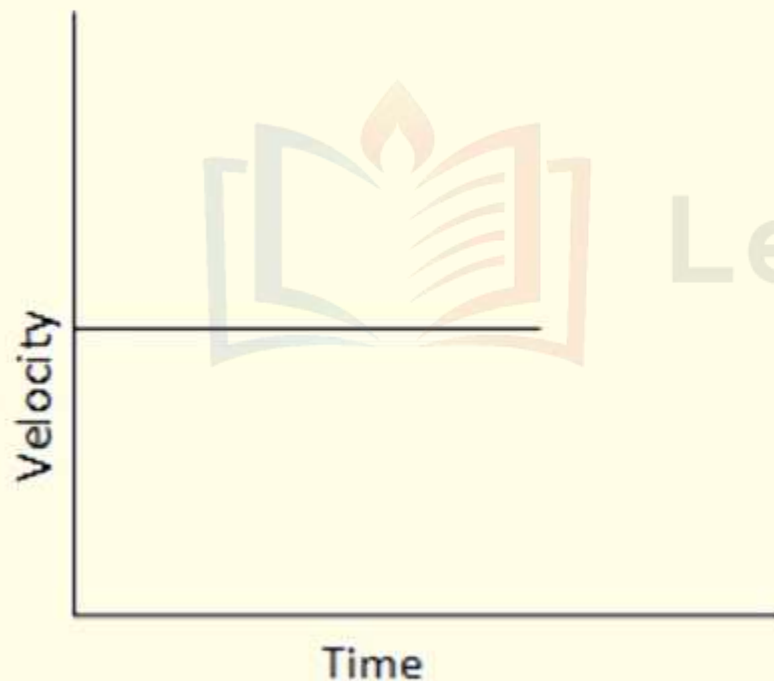
$$\text{SPEED} = \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{AB}{OB}$$



- To calculate speed of the body from distance time graph say at point A first draw a perpendicular AB on time axis and a perpendicular AC on distance axis.
- AB represents the distance travelled by the body in time interval OB.



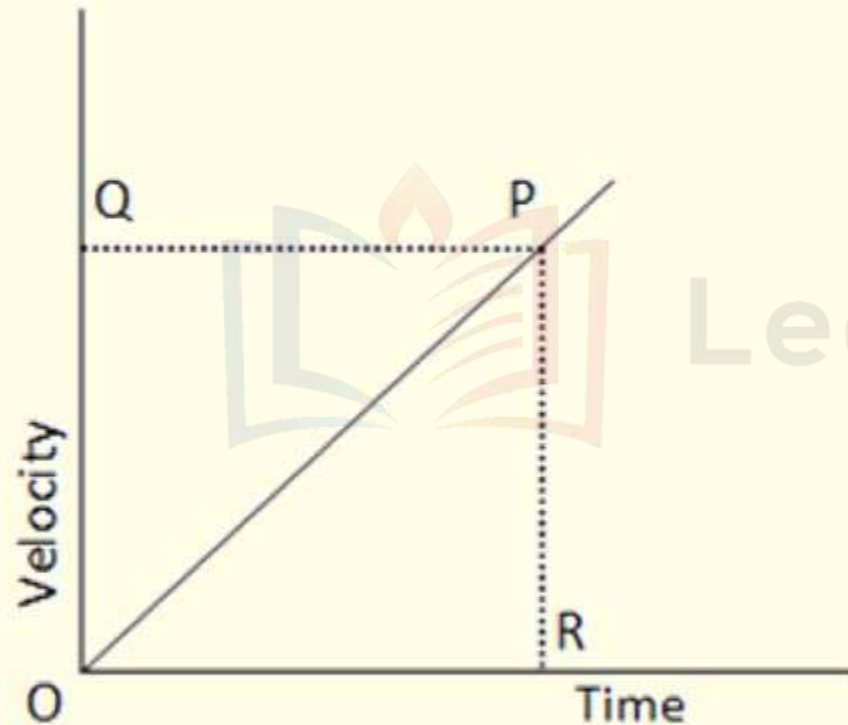
VELOCITY TIME GRAPH



- The variation in velocity with time for an object moving in a straight line can be represented by a velocity-time graph.
- In this graph, time is represented along the x-axis and the velocity is represented along the y-axis.
- If a body moves with a constant velocity then velocity time graph for this body would be straight line parallel to time axis.
- **The product of velocity and time give displacement** of an object moving with uniform velocity.
- **The area enclosed by velocity-time graph and the time axis will be equal to the magnitude of the displacement.**



INFORMATION FROM VELOCITY TIME GRAPH - ACCELERATION

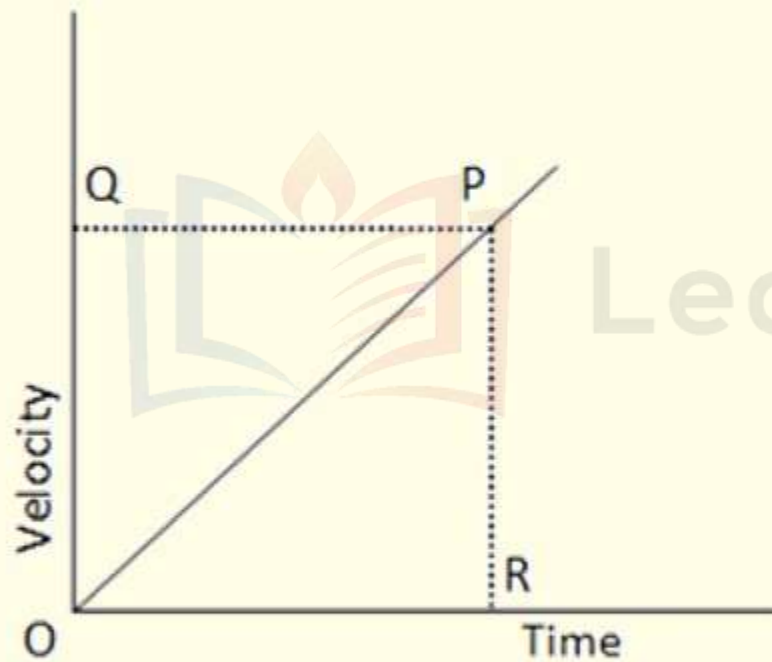


- The slope of the Velocity- time graph gives acceleration.
- For calculating acceleration at time corresponding to point R draw a perpendicular RP from point R.
- Draw QP. PR is the change in velocity when OR is the time taken ;

$$\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time taken}} = \frac{PR}{OR}$$



INFORMATION FROM VELOCITY TIME GRAPH - DISPLACEMENT

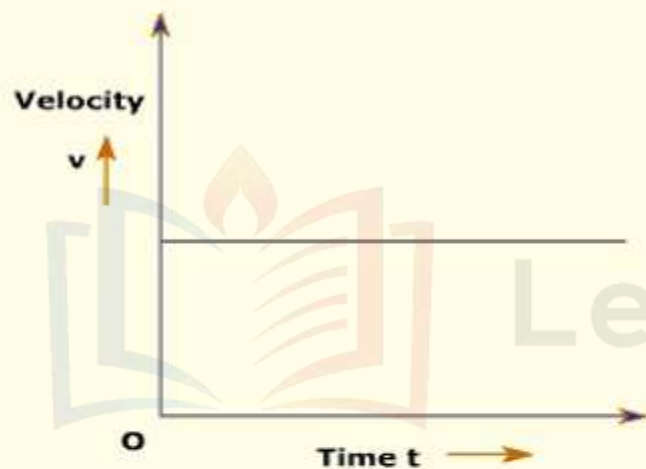


- Displacement travelled will be the area of the triangle OPR.
- As displacement or distance travelled is velocity times the time taken

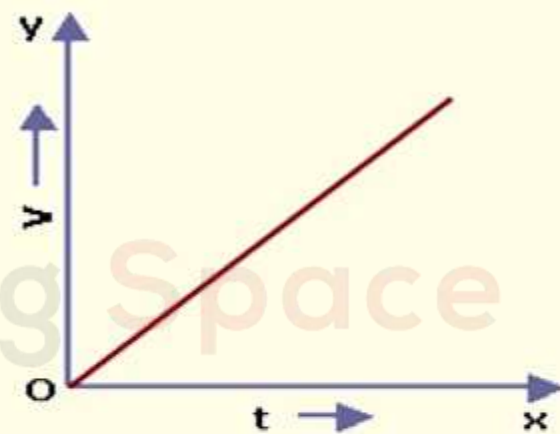
$$D = \frac{1}{2} OQ \times OR = \frac{1}{2} \text{ area of rect. ORPQ}$$



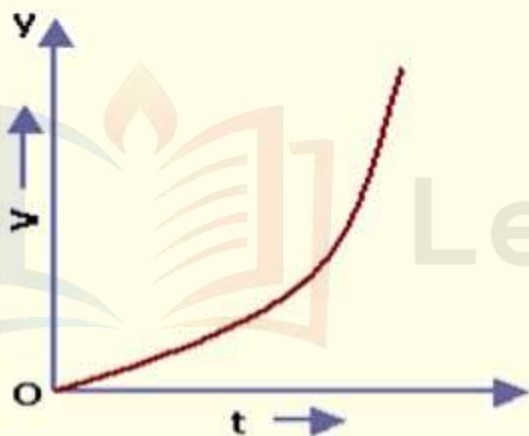
NO ACCELERATION



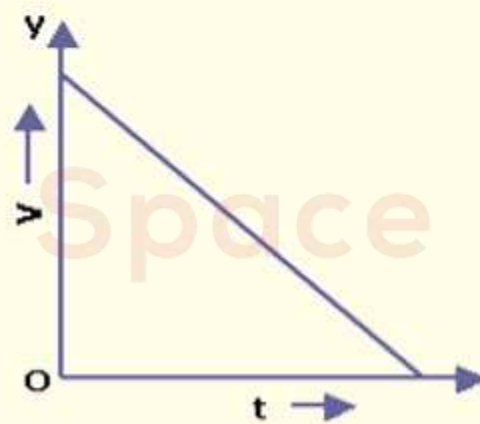
UNIFORM ACCELERATION



NON - UNIFORM ACCELERATION

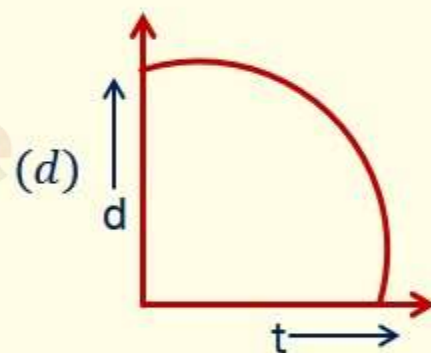
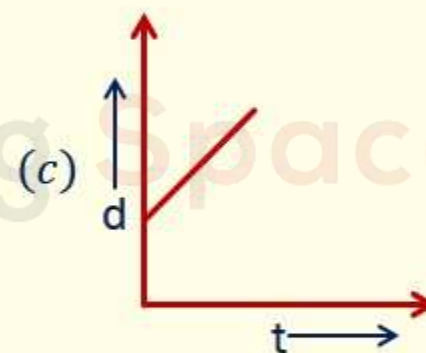
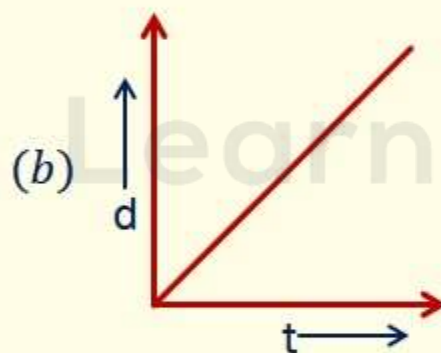
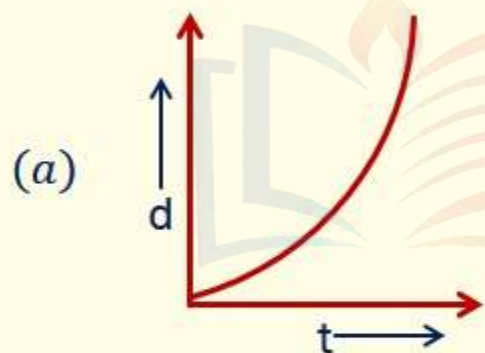


NEGATIVE ACCELERATION



QUESTION 2

The variation of displacement d with time t in the case of a particle falling freely under gravity from rest is correctly shown in graph



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